A protocol for first-week mentor conversations

The first session with a new student is mostly throwaway. Here is how we stopped wasting it.

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The first session a mentor has with a new student is the one that matters most and the one most programs handle worst. We were one of those programs for a long time. We would pair a mentor and a student, hand them a kit, point them at a table, and assume that the relationship would figure itself out from there. Some of them did. Most of them spent the first hour in a kind of polite, low-information drift, and then spent the next three weeks recovering from the bad assumptions they made about each other in that first hour.

This is the protocol we now use for the first week. It is not elegant. It is a list of things mentors are required to do in a specific order, with a specific amount of time allocated to each. The reason it is a list and not a philosophy is that, after enough sessions, we noticed that the mentors who did these things in this order had dramatically better pairings than the mentors who didn't, and we got tired of hoping people would figure it out on their own.

The first ten minutes are about the student, not the robot

The single biggest mistake a mentor can make in the first session is to start with the kit. The kit is in the middle of the table. The kit is exciting. The student is eight, or twelve, and is looking at the kit. Everyone wants to start with the kit. You do not start with the kit.

You start with three questions, in this order. What did you do today before you came here? What is something you built or made recently, anything, doesn't have to be robotics? What did you think you were signing up for when you came to this program?

These are not warm-up questions. They are diagnostic. The first one tells you what kind of day the student has had, which determines how much energy they have for what is about to happen. A kid who came straight from a two-hour math test is not the same kid as one who just had recess. The second question tells you what they consider building. Some students will say a Lego set. Some will say a sandcastle. Some will say a drawing. Some will say nothing, which is also information. The third question tells you what their expectations are, which tells you how much expectation-management you need to do before any actual work begins.

Mentors are required to write down the answers. Not because we audit the notes, but because the act of writing makes you listen, and the answers are reference material for the next ten sessions. A student who told you in week one that they came here because their mom signed them up is a different coaching problem than a student who told you they came here because they want to build a robot dog. Both are workable. You need to know which one you have.

The forty-five-minute build is a test, but the student doesn't know that

After the conversation, the mentor introduces the kit and proposes the same task to every new student, regardless of age or background. Build something that moves. Not anything in particular. Anything. You have forty-five minutes. I will be sitting here. If you ask me a question, I will answer it, but I will not start anything for you.

This sounds casual. It is not. It is the most diagnostic forty-five minutes you will get with this student for months. The mentor is watching for a specific list of things, and is writing them down as they happen. Does the student start by reading the instructions, by dumping the parts on the table, or by asking what to do? Do they reach for wheels first, motors first, or the frame first? When they get stuck, do they look up at the mentor immediately, or do they try something else first? How long can they sit with a problem before they ask for help? When they ask, is the question specific ("this pin won't fit here") or vague ("it's not working")? Do they narrate what they are doing out loud, or do they work in silence?

We give mentors a checklist for this. Twelve items, all yes-or-no, all observed in the first session. The point is not to grade the student. The point is to have a baseline. A student who narrates everything out loud in week one and then stops narrating in week six has changed in a way that is worth understanding. A student who never asks for help in week one is going to need different coaching than a student who asks for help on every step. You cannot calibrate any of this without the first-week baseline, and you cannot get the baseline without giving them an open-ended task and watching what they actually do with it.

The mentor is allowed to answer questions, but not to volunteer. This is the hardest rule to enforce, and the one we drill the most in mentor training. The instinct, especially for engineers, is to see a student doing something inefficiently and to suggest a better way. In the first session, you do not suggest a better way. You let them do it the wrong way, and you write down what the wrong way was, because the wrong way is the most valuable information you will get all hour.

The last five minutes are about the next session, not this one

When forty-five minutes are up, regardless of what the student has built, the mentor stops the build. This is also hard. The student is usually in the middle of something. The temptation is to give them another ten minutes. Don't.

You stop, and you ask three questions. What is one thing you figured out today? What is one thing you got stuck on? If we have another hour next time, what do you want to do with it?

The first question forces the student to articulate a learning, which they have probably not done yet on their own. Most kids, asked this question in the first session, will struggle to answer it. That's fine. The struggle is the point. By session three or four, they will have an answer ready, and that is one of the most useful behavior changes a beginner student goes through. The second question gives the student permission to admit something was hard, which is a permission they are often not used to having. The third question makes the student a participant in planning the next session, which is the single largest factor in whether they show up for it.

The mentor writes all three answers down. The third answer in particular becomes the agenda for session two. If the student said they wanted to make it go faster, the next session is about gears or motor power. If they said they wanted to make it not crash into things, the next session is about sensors. The session is not designed around what the curriculum says comes next. It is designed around what the student said they wanted, in their own words, written down in front of them.

This sounds like a small thing. It is not a small thing. It is the difference between a student who feels like they are being processed through a program and a student who feels like the program is responding to them. The first kind of student leaves when something more interesting comes along. The second kind of student stays.

What goes in the handoff note

The last thing the mentor does, after the student has left, is write a handoff note. Six fields, all short. What did the student tell you about themselves? What did you observe during the build? What did they say they wanted next? What is one thing you would do differently if you started over with them? What is one specific question you would ask their parent if you got the chance? What is the one piece of context the next mentor most needs to know?

The note is for the next mentor in case of a swap, but it is also for the same mentor in week three when they have forgotten half of what happened in week one. Mentors who write these notes consistently have measurably better pairings than mentors who don't. We have looked at this. It is not subtle.

The note takes about four minutes if you are quick and seven if you are thorough. Mentors are required to do it before they leave the building. If they leave without doing it, the note does not get written, because by the time they get home they have lost half the texture of the session. The four to seven minutes happens at the table, while the kit is still in front of them. This is not negotiable. We learned this the hard way.

Why we wrote this down

The reason we wrote this protocol down, instead of leaving it as something experienced mentors knew and inexperienced mentors didn't, is that the gap between a good first session and a bad one was the single largest predictor we could find of whether a student was still in the program three months later. Not the curriculum. Not the kit. Not the mentor's technical strength. The first session.

If you are running a program, the thing I would tell you is that the first hour with a student is not a warm-up. It is the foundation. Spending it on rapport-building chatter, or on jumping into the lesson plan, or on letting the student "explore" without structure, is wasting it. The first hour has specific work to do. Write down what that work is, and make every mentor do it the same way, and you will get a different program out the other side than the one you would have had if you let everyone figure it out on their own.

We let everyone figure it out on their own for two years. The protocol exists because we counted, eventually, how much that cost us.

Growing Up with Robotics is a 501(c)(3) nonprofit. Our full mentor handbook, including the first-week checklist, is available on request.